

بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ

CS458- Data Mining (DM)

Fall 2025

Lecture 3-4: What is Data Mining?

Lets Start!

What is Data Mining and Why do we need it?



**Slides edited from Han and Kamber's online lecture slides*

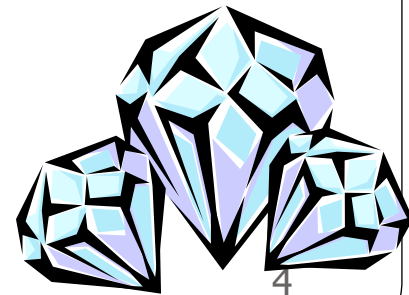
Why Data Mining?

- The Explosive Growth of Data: from terabytes to petabytes
 - Data collection and data availability
 - Automated data collection tools, database systems, Web, computerized society
 - Major sources of abundant data
 - Business: Web, e-commerce, transactions, stocks, ...
 - Science: Remote sensing, bioinformatics, scientific simulation, ...
 - Society and everyone: news, digital cameras, YouTube
- We are drowning in data, but starving for knowledge!
- “Necessity is the mother of invention”—Data mining—Automated analysis of massive data sets

What Is Data Mining?



- Data mining (knowledge discovery from data)
 - Extraction of interesting (non-trivial, implicit, previously unknown and potentially useful) patterns or knowledge from huge amount of data
 - Data mining: a misnomer?
- Alternative names
 - Knowledge discovery (mining) in databases (KDD), knowledge extraction, data/pattern analysis, data archeology, data dredging, information harvesting, business intelligence, etc.
- Watch out: Is everything “data mining”?
 - Simple search and query processing
 - (Deductive) expert systems



Database Processing vs. Data Mining Processing

- Query

- Well defined
- SQL

- Output

- Precise
- Subset of database

- Query

- Poorly defined
- No precise query language

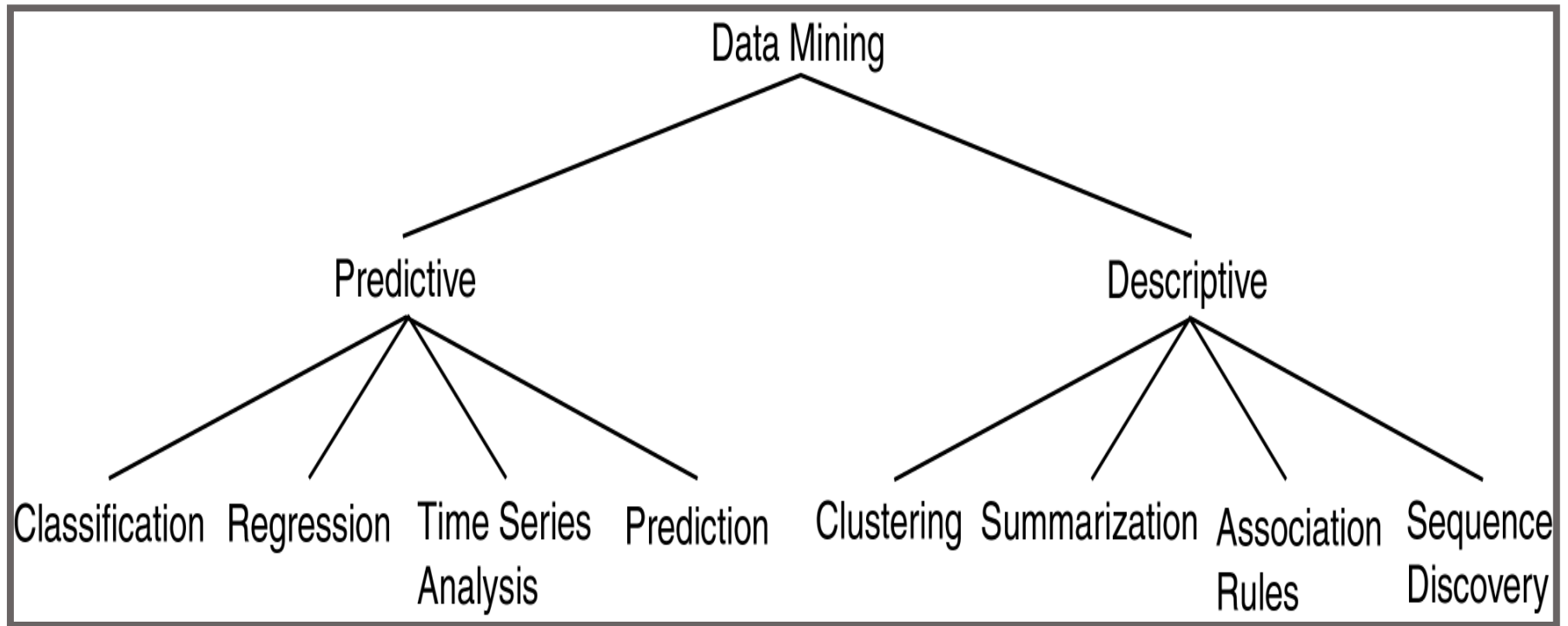
- Output

- Fuzzy
- Not a subset of database

Query Examples

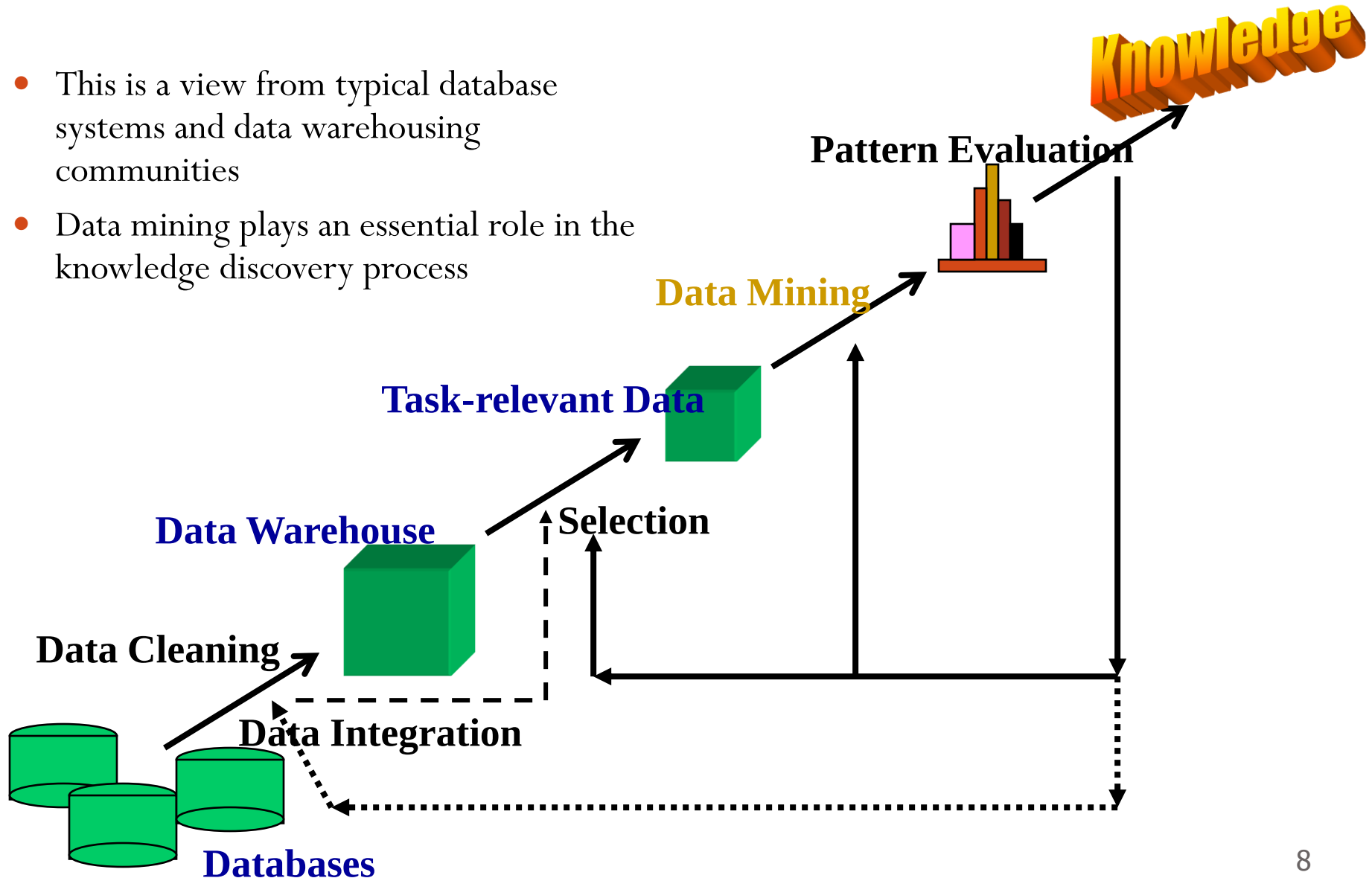
- Database
 - Find all credit applicants with last name of Smith.
 - Identify customers who have purchased more than \$10,000 in the last month.
 - Find all customers who have purchased milk
- Data Mining
 - Find all credit applicants who are low credit risks. (classification)
 - Identify customers with similar buying habits. (Clustering)
 - Find all items which are frequently purchased together. (association rules)

Data Mining Models and Tasks

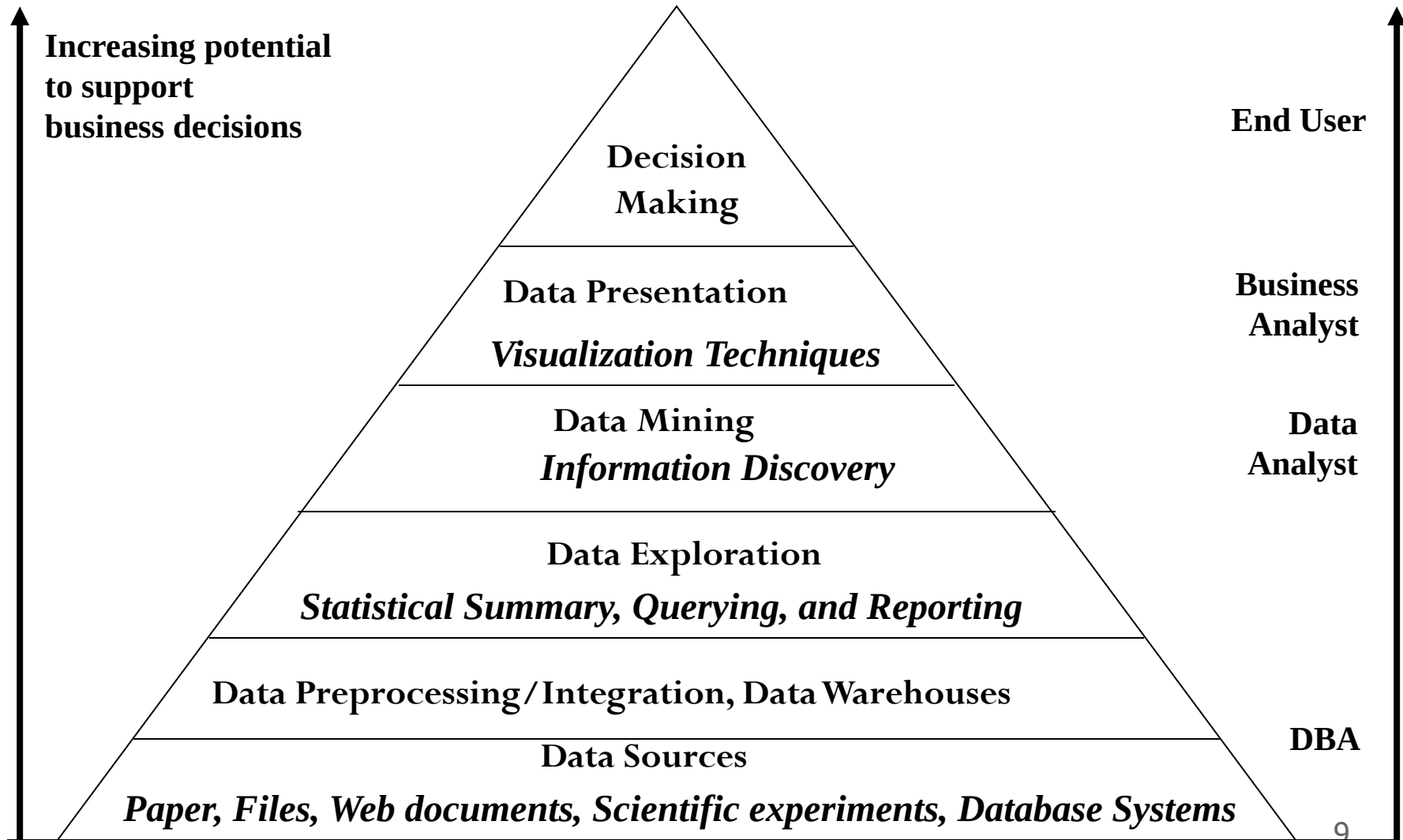


Knowledge Discovery (KDD) Process

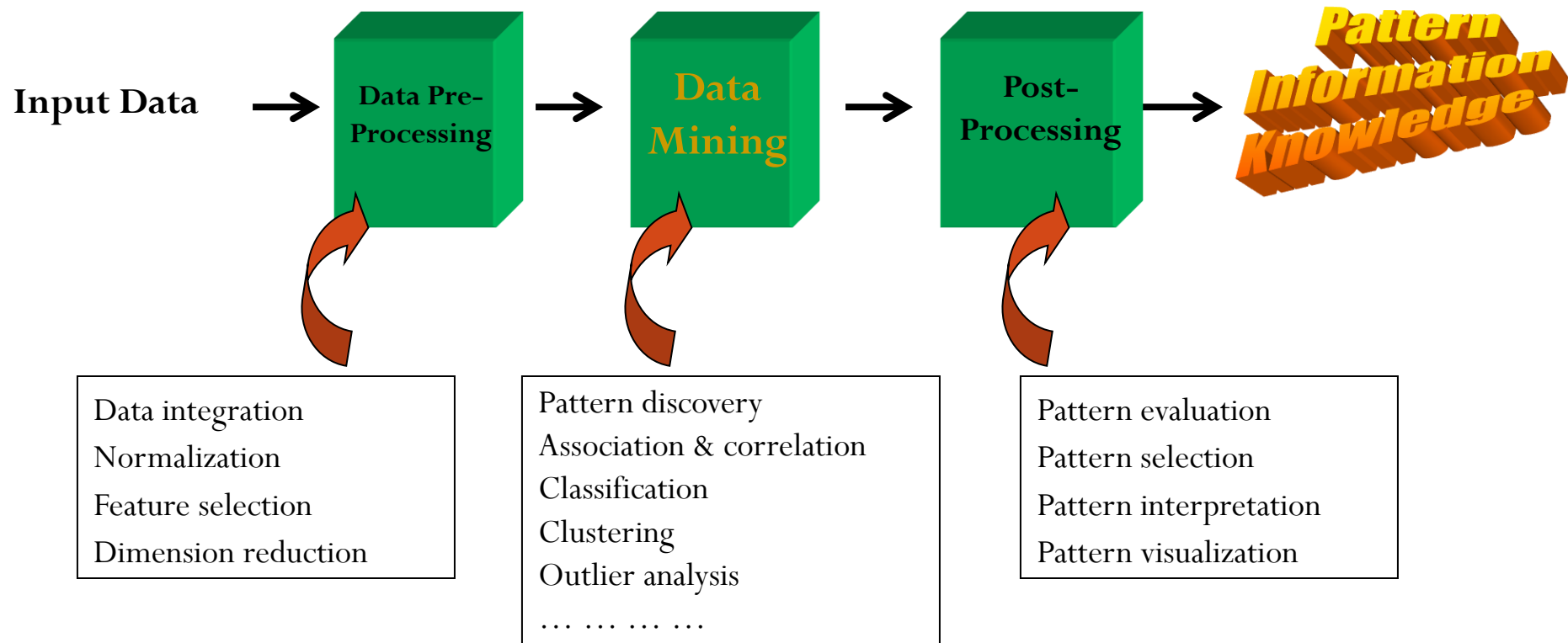
- This is a view from typical database systems and data warehousing communities
- Data mining plays an essential role in the knowledge discovery process



Data Mining in Business Intelligence



KDD Process: A Typical View from ML and Statistics



- This is a view from typical machine learning and statistics communities

Chapter 1. Introduction

- Why Data Mining?
- What Is Data Mining?
- A Multi-Dimensional View of Data Mining 
- What Kinds of Data Can Be Mined?
- What Kinds of Patterns Can Be Mined?
- What Kinds of Technologies Are Used?
- What Kinds of Applications Are Targeted?
- Major Issues in Data Mining
- A Brief History of Data Mining and Data Mining Society
- Summary

What Kinds of Data Can Be Mined?

???*Data* Mining

- Database Data
- Data Warehouse Data
- Transactional Data
- Data Streams
- Graph Data
- Text Data
- Multimedia Data
- ...

Mining Database Data

- The fictitious *AllElectronics* store is used to illustrate concepts throughout the text book. The company is described by the following relation tables:
 - customer, item, employee, and branch.

<i>customer</i>	(<i>cust_ID</i> , <i>name</i> , <i>address</i> , <i>age</i> , <i>occupation</i> , <i>annual_income</i> , <i>credit_information</i> , <i>category</i> , ...)
<i>item</i>	(<i>item_ID</i> , <i>brand</i> , <i>category</i> , <i>type</i> , <i>price</i> , <i>place_made</i> , <i>supplier</i> , <i>cost</i> , ...)
<i>employee</i>	(<i>empl_ID</i> , <i>name</i> , <i>category</i> , <i>group</i> , <i>salary</i> , <i>commission</i> , ...)
<i>branch</i>	(<i>branch_ID</i> , <i>name</i> , <i>address</i> , ...)
<i>purchases</i>	(<i>trans_ID</i> , <i>cust_ID</i> , <i>empl_ID</i> , <i>date</i> , <i>time</i> , <i>method_paid</i> , <i>amount</i>)
<i>items_sold</i>	(<i>trans_ID</i> , <i>item_ID</i> , <i>qty</i>)
<i>works_at</i>	(<i>empl_ID</i> , <i>branch_ID</i>)

Figure 1.5 Relational schema for a relational database, *AllElectronics*.

Mining Database Data

- Relational data can be accessed by database queries
- Through the use of relational queries, you can ask things like
 - Show me a list of all items that were sold in the last quarter.

Mining Database Data

- Relational data can be accessed by database queries
- Through the use of relational queries, you can ask things like
 - Which salesperson had the highest sales?

Mining Database Data

- When **mining** relational databases, we can go further by searching for trends or data patterns
- Analyze customer data to predict the credit risk of new customers based on their income, age, and previous credit information

Mining Database Data

- When **mining** relational databases, we can go further by searching for trends or data patterns
- Items with sales that are far from those expected in comparison with the previous year

Mining Transactional Data

- Each record in a transactional database captures a transaction, such as
 - a customer's purchase
 - a flight booking, or
 - a user's clicks on a web page

Mining Transactional Data

- A transaction typically includes
 - a unique transaction identity number (**trans ID**) and
 - a list of the items making up the transaction, such as
 - the *items purchased in the transaction*

Mining Transactional Data

<i>trans_ID</i>	<i>list_of_item_IDs</i>
T100	I1, I3, I8, I16
T200	I2, I8
...	...

Figure 1.8 Fragment of a transactional database for sales at *AllElectronics*.

Mining Transactional Data

- *Which items sold well together?*

<i>trans_ID</i>	<i>list_of_item_IDs</i>
T100	I1, I3, I8, I16
T200	I2, I8
...	...

Figure 1.8 Fragment of a transactional database for sales at *AllElectronics*.

Mining Transactional Data

- Given the knowledge that
 - **printers** are commonly purchased together with **computers**
- You could offer certain printers at a steep **discount** to customers buying selected computers
 - in the hopes of selling more computers

Chapter 1. Introduction

- Why Data Mining?
- What Is Data Mining?
- A Multi-Dimensional View of Data Mining
- What Kinds of Data Can Be Mined?
- What Kinds of Patterns Can Be Mined? 
- What Kinds of Technologies Are Used?
- What Kinds of Applications Are Targeted?
- Major Issues in Data Mining
- A Brief History of Data Mining and Data Mining Society
- Summary

Data Mining Functionalities

Data Mining Functionalities

- Characterization
- Discrimination
- Association and Correlation Analysis (**Ch. 6**)
- Classification (**Ch. 8**)
- Regression
- Clustering (**Ch. 10**)
- Outlier Analysis

Data Characterization

- *Summarize the characteristics of customers who spend more than \$5000 a year at AllElectronics.*

Data Characterization

- *Summarize the characteristics of customers who spend more than \$5000 a year at AllElectronics.*
- The result is a general profile of these customers, such as that they are
 - **40 to 50 years old,**
 - **employed, and**
 - **have excellent credit ratings.**

Data Discrimination

- *Compare two groups of customers*
 - Those who shop for computer products regularly (e.g., more than twice a month)
 - Those who rarely shop for such products (e.g., less than three times a year)

Data Discrimination

- The resulting description provides a general comparative profile of these customers, such as that
 - 80% of the customers who frequently purchase computer products are between 20 and 40 years old and have a university education
 - Whereas 60% of the customers who infrequently buy such products are either seniors or youths, and have no university degree

Frequent Patterns, Association and Correlation Analysis

- Frequent patterns (or frequent itemsets)
 - What items are frequently purchased together on DARAZ?
- A typical association rule
 - Butter, Bread $\rightarrow$ Milk [40%, 100%] (support, confidence)
- How to mine such patterns and rules efficiently in large datasets?

Association Rule Example

- $X \rightarrow Y$
- Butter, Bread $\rightarrow$ Milk
- [40%, 100%] (support, confidence)

Example dataset with 4 items and 5 transactions

Transaction Id	Milk	Bread	Butter	Soap
1	1	1	0	0
2	1	1	1	0
3	0	0	0	1
4	1	1	1	0
5	0	1	0	0

- Support
 - the proportion of transactions in the data set which contain the itemset
 - 2/5
- Confidence
 - $\text{conf}(X \Rightarrow Y) = \text{supp}(X \cup Y) / \text{supp}(X)$
 - 2/2
 - What about Bread $\rightarrow$ Milk ?

Association Rule Example

- $X \rightarrow Y$
- Chicken, Clothes $\rightarrow$ Milk ?

Example dataset with 4 items and 5 transactions

Transaction Id	Items bought
1	Beef, Chicken, Milk
2	Beef, Cheese
3	Cheese, Shoes
4	Beef, Chicken, Cheese
5	Beef, Chicken, Cheese, Clothes, Milk
6	Chicken, Clothes, Milk
7	Chicken, Milk, Clothes

- Support
 - the proportion of transactions in the data set which contain the itemset
- Confidence
 - What about Clothes $\rightarrow$ Milk, Chicken ? $\text{conf}(X \Rightarrow Y) = \text{supp}(X \cup Y) / \text{supp}(X)$

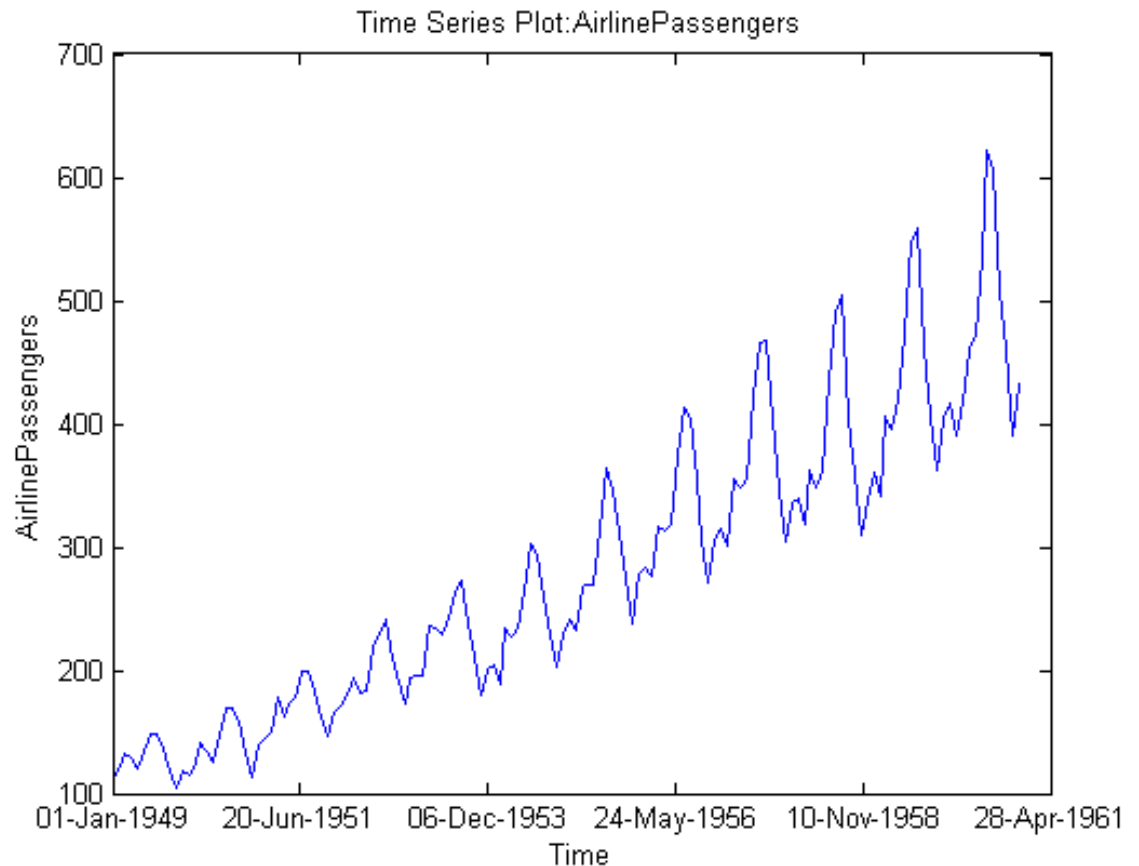
Classification

- Classification and **label prediction**
 - Construct models (functions) based on some training examples
 - Describe and distinguish classes or concepts for future prediction
 - Predict some unknown class labels
- Typical methods
 - Decision trees, naïve Bayesian classification, support vector machines, neural networks, rule-based classification, pattern-based classification, logistic regression, ...
- Typical applications:
 - Credit card fraud detection, direct marketing, disease identification, ...

Regression

classification predicts categorical (discrete, unordered) labels, **regression** models continuous-valued functions.

Numeric Prediction



Cluster Analysis

- Unsupervised learning (i.e., Class label is unknown)
- Group data to form new categories (i.e., clusters), e.g., cluster houses to find distribution patterns
- Principle: Maximizing intra-class similarity & minimizing interclass similarity
- Many methods and applications
 - Market segmentation
 - Recommender systems
 - Social network analysis
 - Education

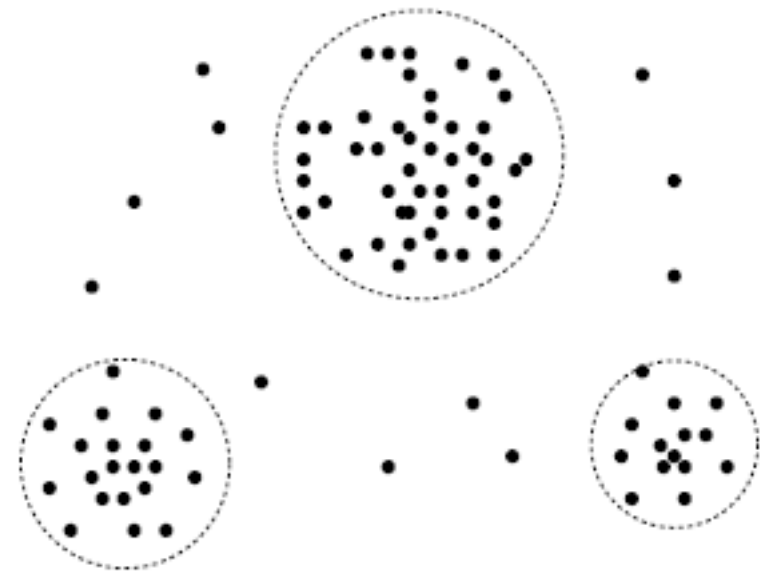
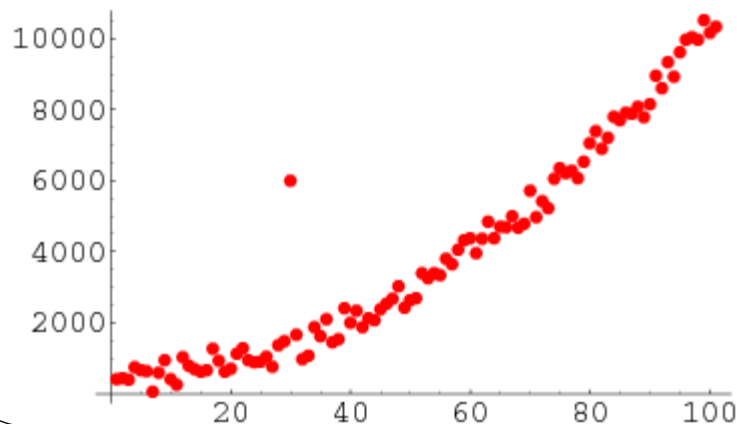


Figure 1.10 A 2-D plot of customer data with respect to customer locations in a city, showing three data clusters.

Outlier Analysis

- Outlier analysis
 - Outlier: A data object that does not comply with the general behavior of the data
 - Noise or exception? — One person's garbage could be another person's treasure
 - Methods: by product of clustering or regression analysis, ...
 - Useful in fraud detection, rare events analysis



Class Activity

- Discuss whether or not each of the following activities is a data mining task.
- A) Dividing the customers of a company according to their gender.
- No. This is a simple database query.
- B) Dividing the previous customers of a company according to their profitability.
- No. This is an accounting calculation, followed by the application of a threshold. However, predicting the profitability of a new customer would be data mining.

Data Mining yes/no?

- (c) Computing the total sales of a company.
- No. Again, this is simple accounting.
- (d) Sorting a student database based on student identification numbers.
- No. Again, this is a simple database query.
- (e) Predicting the outcomes of tossing a (fair) pair of dice.
- No. Since the die is fair, this is a probability calculation. If the die were not fair, and we needed to estimate the probabilities of each outcome from the data, then this is more like the problems considered by data mining. However, in this specific case, solutions to this problem were developed by mathematicians a long time ago, and thus, we wouldn't consider it to be data mining.

Data Mining yes/no?

- (f)Predicting the future stock price of a company using historical records.
- Yes. We would attempt to create a model that can predict the continuous value of the stock price. This is an example of the area of data mining known as predictive modelling. We could use regression for this modelling, although researchers in many fields have developed a wide variety of techniques for predicting time series.

Data Mining yes/no?

- (g) Monitoring the heart rate of a patient for abnormalities.
- Yes. We would build a model of the normal behavior of heart rate and raise an alarm when an unusual heart behavior occurred. This would involve the area of data mining known as anomaly detection. This could also be considered as a classification problem if we had examples of both normal and abnormal heart behavior.

Data Mining yes/no?

- (h) Monitoring seismic waves for earthquake activities.
- Yes. In this case, we would build a model of different types of seismic wave behavior associated with earthquake activities and raise an alarm when one of these different types of seismic activity was observed. This is an example of the area of data mining known as classification.
- Extracting the frequencies of a sound wave.
- No. This is signal processing.

Data Mining and Data Privacy

- For each of the following data sets, explain whether or not data privacy is an important issue.
- (a) Census data collected from 2000–2018.
- No
- (b) IP addresses and visit times of Web users who visit your Website.
- Yes
- (c) Patients' information and medical records from a hospital.
- Yes
- (d) Names and addresses of people from the telephone book.
- No
- (e) Names and email addresses collected from the Web.
- No

Homework

- Read chapter 1 of your Data Mining text book.
- Identify similarities and differences in the following
 - Database and Datawarehouse
 - Discrimination and Classification
 - Characterization and Clustering
 - Classification and Regression
 - Classification and Clustering
 - Descriptive and predictive data mining